

Question 2 of 5

a)

$$f(x) = \frac{8}{x^3} \quad x > 2$$

$$g(y) = 2y \quad 0 < y < 1$$

$$E(X) = \int_2^{\infty} \frac{8}{x^3} dx$$

$$= 8 \int_2^{\infty} \frac{1}{x^2} dx$$

$$= 8 \left[-\frac{1}{x} \right]_2^{\infty}$$

$$= 8 \left(-0 + \frac{1}{2} \right)$$

$$= 4$$

(2)

$$E(Y) = \int_0^1 y \cdot 2y \, dy = 2 \int_0^1 y^2 \, dy$$

$$= 2 \left[\frac{y^3}{3} \right]_0^1$$

$$= \frac{2}{3}$$

(2)

$$E(XY) = E(X)E(Y) = 4 \times \frac{2}{3} = \frac{8}{3}$$

as X, Y independent

(2)

b.

$$P(\text{selecting an adult of 40 year with cancer}) = 0.05$$

$$P(\text{correct diagnose} / \text{with cancer}) = 0.78$$

$$P(\text{incorrect diagnosing} / \text{without cancer}) = 0.06$$

$$~~P(\text{diagnosed as having cancer} / \text{40 year adult}) = ?~~$$

$$P(\text{diagnosed as having cancer})$$

$$= P(\text{diagnosed as having cancer} \cap \text{with cancer}) + P(\text{diagnosed as having cancer} \cap \text{without cancer})$$

$$= P(\text{diagnosed as having cancer} / \text{with cancer}) P(\text{with cancer}) + P(\text{diagnosed as having cancer} / \text{without cancer}) P(\text{without cancer})$$

$$= (0.78) \times (0.05) + (0.06) (1 - 0.05)$$

$$= \underline{0.096}$$

(7)

c/.

$$P(X = -2) = \frac{1}{6}$$

$$P(X = -1) = \frac{1}{3}$$

$$P(X = 1) = \frac{1}{4}$$

$$P(X = 2) = \frac{1}{4}$$

(4)

$$P(|X| \leq 1) = P(-1 \leq X \leq 1)$$

$$= P(X = -1) + P(X = 1)$$

$$= \frac{1}{3} + \frac{1}{4}$$

$$= \frac{7}{12}$$

(3)